

REPORT

ARTIFICIAL INTELLIGENCE (CSA1717)

Assessment Tool 1 – Analytical Problem-Solving Report

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1. INTRODUCTION

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks requiring human intelligence, such as learning, reasoning, decision-making, and problem-solving. AI systems use intelligent agents and search algorithms to solve real-world problems efficiently. This assessment demonstrates the application of AI concepts including intelligent agents, state-space search, constraint satisfaction, and optimal path-finding through five analytical problems

2.WATER JUG PROBLEM

Solution

The Water Jug Problem is solved using state-space search, where each state represents the amount of water in the two jugs. The objective is to obtain exactly 2 gallons of water in the 4-gallon jug by performing valid operations.

Allowed Operations

- Fill a jug completely.
- Empty a jug.
- Pour water from one jug to another until one jug is full or the other is empty.

Solution steps:

STEP	OPERATION	4 – GALLON	3 – GALLON
1	Fill the 3-gallon jug	0	3
2	Pour into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour water into the 4-gallon jug until it becomes full	4	2
5	Empty the 4-gallon jug	0	2
6	Transfer the remaining 2 gallons to the 4-gallon jug	2	0

Goal Achieved: The 4-gallon jug contains exactly 2 gallons.

Result

The goal state **(2,0)** is reached successfully.

Conclusion

This problem demonstrates how an AI agent explores different states systematically to achieve a goal using an uninformed search strategy.

3.MARS ROVER

Solution

The Mars Rover is an autonomous intelligent agent designed to explore Mars without direct human control.

Percepts

- Camera images
- Rock and soil samples
- Obstacle information
- Temperature
- Battery level
- Position and terrain

Operating Environment

- Partially observable
- Dynamic
- Sequential
- Continuous
- Unknown
- Remote environment

Actions

- Move forward and backward
- Turn left and right

- Capture images
- Collect rock samples
- Analyze soil
- Avoid obstacles
- Send data to Earth

Performance Measure

- Collect maximum scientific data
- Navigate safely
- Minimize energy consumption
- Complete mission successfully

Suitable Agent

Utility-Based Agent

Justification

A utility-based agent evaluates several possible actions and selects the one that maximizes mission success while minimizing energy consumption and risk. It performs efficiently in uncertain and changing environments.

Conclusion

The Mars Rover is an excellent example of an intelligent autonomous agent that uses AI to make decisions in real time.

4.8-QUEENS PROBLEM

Solution

The 8-Queens Problem is a Constraint Satisfaction Problem (CSP). The objective is to place eight queens on an 8×8 chessboard so that no queen attacks another queen.

Initial State

- Empty chessboard

State Space

- All possible arrangements of queens

Operators

- Place a queen
- Remove a queen
- Move a queen

Goal Test

- Eight queens are placed.
- No two queens share the same row, column, or diagonal.

Path Cost

- Each queen placement is treated as one step.

Search Strategy

- Backtracking Search
- Depth-First Search (DFS)

Advantages

- Eliminates invalid configurations early.
- Efficiently explores the search space.
- Guarantees a valid solution if one exists.

Conclusion

The 8-Queens Problem demonstrates AI search techniques and constraint satisfaction methods used in scheduling, planning, and optimization.

5. OLA CAB BOOKING

Solution

The OLA Cab Booking application uses an intelligent agent to recommend the best cab according to the customer's travel needs.

Agent Type

Goal-Based Agent

Working Process

1. User enters pickup location.

2. User enters destination.
3. System checks available cabs.
4. User selects a preferred cab type.
5. System calculates fare and estimated arrival time.
6. Booking is confirmed.

Pseudocode

START

Input Pickup Location

Input Destination

Display Available Cabs

Select Cab Type

Calculate Fare

Confirm Booking

END

Advantages

- Fast booking process.
- Suggests suitable cab options.
- Improves customer satisfaction.
- Reduces waiting time.

Conclusion

A goal-based agent makes decisions based on the user's objective and selects the most appropriate cab to complete the journey efficiently.

6.UNIFORM COST SEARCH (UCS) – SOLUTION

Solution

Uniform Cost Search (UCS) is an uninformed search algorithm that always expands the node with the lowest cumulative path cost. It guarantees the optimal solution when all edge costs are non-negative.

STEP 1: GRAPH DETAILS:

Edges and Costs:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

STEP 2: UCS SEARCH PROCESS:

Step **Expanded Node** **Path Cost** **Frontier (Queue)**

Path Cost

$S \rightarrow A \rightarrow C \rightarrow G$ 4

$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$ 6

$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$ 10

$S \rightarrow G$ 12

STEP 3: LEAST-COST PATH

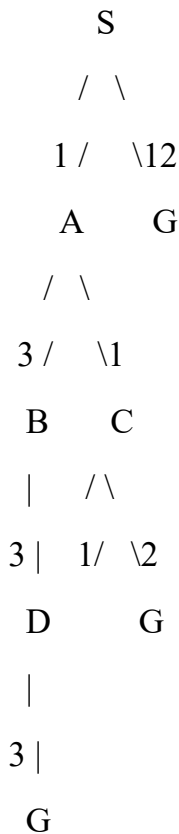
The optimal path found by UCS is:

$S \rightarrow A \rightarrow C \rightarrow G$

TOTAL COST

$$1 + 1 + 2 = 4$$

STEP 4: SEARCH TREE:



Optimal Path: $S \rightarrow A \rightarrow C \rightarrow G$

Total Cost = 4

Conclusion

Uniform Cost Search is an efficient algorithm for solving shortest-path problems because it always selects the path with the minimum accumulated cost.

7. CONCLUSION

This assessment demonstrates the practical application of Artificial Intelligence techniques in solving different categories of problems. The Water Jug Problem illustrates state-space search, the Mars Rover highlights intelligent agent design, the 8-Queens Problem demonstrates constraint satisfaction and search strategies, the OLA Cab application represents goal-based decision making, and Uniform Cost Search finds the optimal path in a weighted graph. These examples show how AI algorithms improve decision-making, optimization, and automation across real-world domains.

8. REFERENCES

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